


7

NUMBER AND ALGEBRA

# DECIMALS

Before decimal currency was introduced in Australia in 1966, our money was based on the British pound, shilling and penny. 1 pound = 20 shillings and 1 shilling = 12 pence. Before the metric system was adopted in Australia in 1974, we used the imperial system of measurement. For example, 1 mile = 1760 yards, 1 yard = 3 feet, 1 foot = 12 inches. This meant that 60 years ago the calculations and conversions were more complicated. Today, our currency and measurement units follow a decimal system using powers of 10. For example, 1 L = 1000 mL, 1 m = 100 cm, 1 cm = 10 mm.

With the metric system and computer technology, decimals have become more commonly used than fractions to describe parts of a whole.




## Chapter outline

| | Working mathematically | | | | |
|---|---|---|---|---|---|
| **7.01** Ordering decimals | U | F | PS | R | C |
| **7.02** Decimals and fractions | U | F | | | C |
| **7.03** Adding and subtracting decimals | U | F | PS | R | C |
| **7.04** Multiplying and dividing decimals by powers of 10 | U | F | | R | |
| **7.05** Multiplying decimals by estimating | U | F | | | |
| **7.06** Multiplying decimals | U | F | PS | | C |
| **7.07** Dividing decimals by whole numbers | U | F | | R | |
| **7.08** Dividing decimals | U | F | | R | C |
| **7.09** Terminating and recurring decimals | U | F | | R | C |
| **7.10** Rounding decimals | U | F | | R | C |
| **7.11** Decimal problems | U | F | PS | R | C |

## Wordbank

**ascending** Increasing from smallest to largest

**decimal places** The places after the decimal point in a number

**recurring decimal** A decimal that has one or more digits that repeat endlessly

**round** To write a number to a given number of places

**tenth** The fraction $\frac{1}{10}$ or the decimal 0.1, the first decimal plac

**terminating decimal** A decimal that is not recurring, but comes to an end

**Quiz**
Wordbank 7

*Note:* Multiplying and dividing decimals (including by whole numbers) using written methods is no longer in the syllabus. Teachers can use judgement to teach calculator methods only.

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

### Videos (8):

**7.01** Naming decimals
**7.03** Adding and subtracting decimals
**7.04** Multiplying and dividing decimals by powers of 10
**7.06** Multiplying decimals
**7.07** Dividing decimals
**7.09** Converting fractions to decimals • Recurring decimals
**7.10** Rounding decimals

### *Twig* video (1):

**7.01** Decimal places: Photofinish

### *PhET* interactive (1):

**7.06** Area Model Decimals

### Quizzes (5):

- Wordbank 7
- SkillCheck 7
- Mental skills 7
- Language of maths 7
- Test yourself 7

### Skillsheets (3):

**SkillCheck** Decimals • Multiplying by 10, 100, 1000
**7.02, 7.09** Fractions and decimals

### Worksheets (12):

**7.01** Decimals wall • Dewey decimals • Decimals 1 • Decimals 10
**7.06** Shopping and change • Decimal cards • Where's the point?
**7.08** Decimals writing activity • Decimals 7 • Decimal number grids
**7.11** Shopping and change • Calculating change
**Mind map** Decimals

### Puzzles (6):

**7.02** Decimals squaresaw 1
**7.06** Which decimals? • Multiplication estimation game
**7.08** Operations with decimals
**7.09** Decimals squaresaw 2
**Language of maths** Decimals crossword

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

## In this chapter you will:

- compare and order decimals
- convert between fractions and decimals
- solve problems involving adding and subtracting decimals
- multiply and divide decimals by powers of 10
- solve problems involving multiplying and dividing decimals
- convert a fraction to a terminating or recurring decimal
- round a decimal to a specified number of decimal places
- solve problems involving decimals

## SkillCheck ANSWERS ON P. 584

**1** Copy this place value table so that it has 8 blank rows.

| hundreds | tens | units | | tenths | hundredths | thousandths |
|---|---|---|---|---|---|---|
| | | | . | | | |
| | | | . | | | |
| | | | . | | | |

Write each decimal in the table, with the digits in their correct columns.

**a** 14.82 **b** 6.014 **c** 931.02 **d** 70.8

**e** 0.375 **f** 0.19 **g** 8.592 **h** 715.3

**2** Write the value of the digit 4 in each decimal.

**a** 431.70 **b** 52.407 **c** 761.84

**d** 114.65 **e** 2.734 **f** 907.481

**3** Simplify each fraction.

**a** $\frac{18}{100}$ **b** $\frac{4}{10}$ **c** $\frac{350}{1000}$ **d** $\frac{24}{100}$

**4** List each set of numbers in ascending order.

**a** 21, 32, 34, 30, 20 **b** 213, 216, 234, 206, 230

**5** Evaluate each expression.

**a** $32 \times 10$ **b** $578 \times 100$ **c** $325 \times 1000$

**d** $400 \times 10$ **e** $400 \div 10$ **f** $1200 \div 100$

**g** $1\,000\,000 \div 1000$ **h** $81\,000 \div 100$ **i** $640 \div 10$

**6** List each set of numbers in descending order.

**a** 44, 39, 42, 45, 38, 40 **b** 505, 556, 513, 520, 549, 522

**7** Round:

**a** 27 to the nearest ten **b** 752 to the nearest hundred

**c** 9079 to the nearest thousand **d** 16 837 to the nearest thousand

**8** Estimate the value of each expression.

**a** $140 + 32 + 381$ **b** $432 + 45 + 341 + 7$ **c** $746 - 29$

**d** $596 - 338$ **e** $1138 - 374$ **f** $58 \times 3$

**g** $126 \times 25$ **h** $5920 \div 8$ **i** $2233 \div 7$

**9** Evaluate each expression in question **8** without using a calculator.

**Quiz**
SkillCheck 7

**Skillsheets**
Decimals

Multiplying by 10, 100, 1000

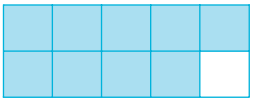
# 7.01 Ordering decimals

**Videos**
Naming decimals
Decimal places: Photofinish

**Skillsheet**
Decimals

**Worksheets**
Decimals wall
Dewey decimals
Decimals 1
Decimals 10

A **decimal** is a special type of fraction where the denominator is a **power of 10** such as 10, 100 or 1000. This shape has been divided into 10 equal parts, of which 9 have been shaded.



- As a fraction, $\frac{9}{10}$ or 'nine-tenths' of the shape is shaded.
- As a decimal, 0.9 or 'zero-point-nine' of the shape is shaded.

You are familiar with numbers that have decimal points, for example:

- money $532.81
- measurements 6.2 km.

A decimal is an example of a **rational number** because it can be written as a fraction $\frac{a}{b}$ where $a$ and $b$ are integers but $b \neq 0$.

The position of a digit in a number shows its size. This is called the **place value**.

In a decimal, the digits after the decimal point indicate a part of a whole. For example, the meaning of the decimal 532.81 is shown on the place value table below:

| hundreds<br>100 | tens<br>10 | units<br>1 | | tenths<br>$\frac{1}{10}$ | hundredths<br>$\frac{1}{100}$ |
|---|---|---|---|---|---|
| 5 | 3 | 2 | . | 8 | 1 |

↑
decimal point

$532.81 = 5 \times 100 + 3 \times 10 + 2 \times 1 + 8 \times \frac{1}{10} + 1 \times \frac{1}{100}$.

## Decimal places

The number of digits after the decimal point tells us the number of **decimal places** in a decimal.

## Example 1

How many decimal places are there in:

**a** 3.6567? **b** 15.801?

### Solution

**a** 3.6567 has 4 decimal places.
1234

**b** 15.801 has 3 decimal places.
123

## Example 2

Arrange these decimals in ascending order: 67.41, 67.14, 6.714, 67.04.

### Solution

To compare decimals more easily, place them in a column with the decimal points in a vertical column and make them all have the same number of decimal places by inserting 0s at the end.

For example, 67.41 becomes 67.410.

67.410
67.140
6.714
67.040

All numbers now have 3 decimal places.

From smallest to largest: 6.714, 67.040, 67.140, 67.410

In ascending order: **6.714, 67.04, 67.14, 67.41**

## Example 3

Arrange these numbers in descending order: 0.5, 0.08, 1.7, 0.85.

### Solution

To compare decimals more easily, place them in a column with the decimal points in a vertical column and make them all have the same number of decimal places by inserting 0s at the end.

0.5 becomes 0.50 and 1.7 becomes 1.70.

0.50
0.08
1.70
0.85

All numbers now have 2 decimal places.

From largest to smallest: 1.70, 0.85, 0.50, 0.08

In descending order: **1.7, 0.85, 0.5, 0.08**

**EXERCISE 7.01** ANSWERS ON P. 585

## Ordering decimals U F PS R C

**1** How many decimal places does each number have?

| | | | | | |
|---|---|---|---|---|---|
| **a** | 1.65 | **b** | 3.881 | **c** | 15.3062 |
| **d** | 0.005 | **e** | 7.045 73 | **f** | 814.3 |
| **g** | 9.597 684 | **h** | 203.602 39 | **i** | 0.0424 706 |

**2** Which of these is the smallest decimal? Select the correct answer **A**, **B**, **C** or **D**.

**A** 1.07 **B** 1.7 **C** 1.077 **D** 1.77



EXAMPLE 1

EXAMPLE 2

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**3** Arrange each set of numbers in ascending order. R

**a** 43.89, 56.324, 9.998, 80.879, 400, 23.89, 56.314

**b** 0.568, 0.684, 0.099, 1.002, 0.586, 5.608, 0.0586

**c** 1.23, 0.891, 1.814, 0.222, 7.007, 0.89

**d** 0.5, 0.05, 0.005

**e** 3.441, 3.404, 3.4, 3.44, 3.004, 3.044

**f** 0.2, 0.202, 0.22, 0.022

EXAMPLE 3

**4** Which of these is the largest decimal? Select **A**, **B**, **C** or **D**.

**A** 2.509 **B** 2.059 **C** 2.59 **D** 2.05

**5** Arrange each set of numbers in descending order. R

**a** 570.25, 125.63, 0.9899, 4000.99, 1256.3, 400.099

**b** 5.37, 6.539, 5.639, 5.367, 3.659, 3.66, 5.369

**c** 1.6, 1.61, 1.599, 1.601, 1.509

**d** 6, 0.06, 0.6, 6.6

**e** 0.7, 0.07, 0.707, 0.77, 0.007, 7.07

**f** 0.4004, 0.044, 0.404, 0.44

**6** Insert < or > to make each statement true. R C

**a** 0.2 ___ 0.25 **b** 0.731 ___ 0.73

**c** 0.035 ___ 0.305 **d** 0.007 ___ 0.070

**e** 1.59 ___ 1.059 **f** 0.099 ___ 0.99

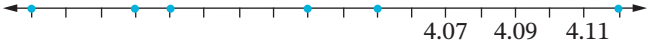
**g** 44.44 ___ 4.444 **h** 0.7932 ___ 0.7239

**7** Copy each number line and write the values of the points marked with dots. R

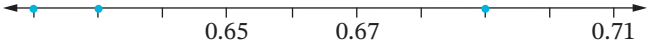
**a**



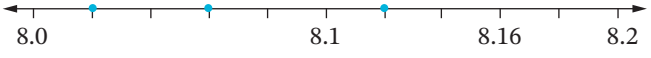

**b**




**c**




**d**




**e**




**f**




**g**




**8** Write a decimal that is: R C

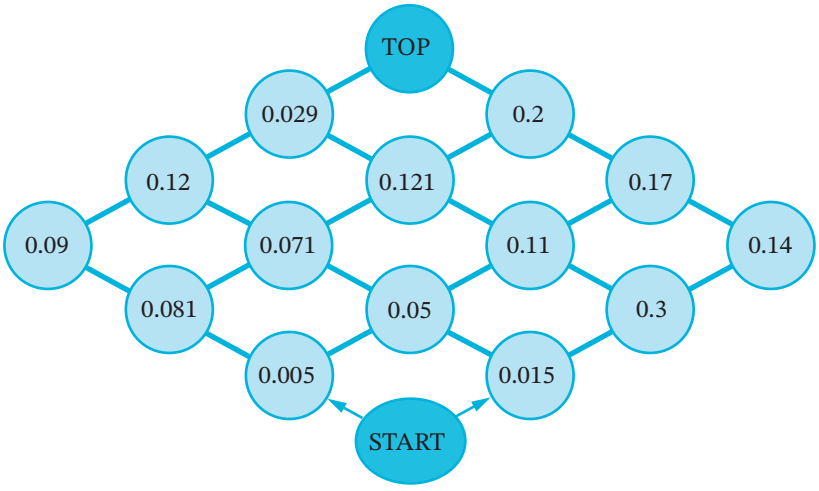
**a** less than 1.3 **b** greater than 2.07

**c** between 8.25 and 8.5 **d** between 6.401 and 6.41

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**9** Find a path from START to TOP. You can make your move from START in any direction, but then you can only move to a larger decimal. Try to find the shortest path, then try to find the longest path. PS R




**10** A square has an area of 2 m$^2$.

**a** Between which 2 whole numbers does the side length lie?

**b** Between which 2 decimals does the side length lie, to one decimal place?

**c** Show the position of the side length on a number line.

## Investigation

### Comparing heights

**1** Use the clues below to find each girl's height.

- Mandy is taller than Sarah.
- Sarah is shorter than Amal.
- Kelly is taller than Sarah but shorter than Mandy.
- Mandy is not the tallest.

The heights of the girls are 168.5 cm, 166.3 cm, 164.2 cm and 160.7 cm.



Left to right: Shutterstock.com/wong yu liang, Shutterstock.com/pikselstock, iStock.com/aldomurillo, Shutterstock.com/Gino Santa Maria

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**2** Use the clues below to find each boy's height.

- Steve is 164.7 cm tall.
- Mike is 14.7 cm taller than Milof.
- Steve is 3.9 cm shorter than Milof.
- Liong is 1.6 cm taller than Mike.






Left to right: iStock.com/jaroon, Shutterstock.com/sheff, Shutterstock.com/Viorel Sima, Shutterstock.com/East

**3** Use the clues below to find each student's height.

- Yoko is 15.1 cm taller than Peter.
- Jade is 13.7 cm shorter than Yoko.
- Karl is 20.6 cm taller than Jade.
- Peter is 163 cm tall.






Left to right: Shutterstock.com/pikselstock, Shutterstock.com/5 second Studio, Shutterstock.com/Maridav, Shutterstock.com/pikselstock

**4** Create your own problem using 4 students from your class. Estimate their heights and height differences. Write a set of clues. (Don't forget to change their names!)

# Decimals and fractions 7.02

## Example 4

Convert each decimal to a simplified fraction.

**a** 0.09 **b** 0.274 **c** 1.45

### Solution

**a** $0.09 = \frac{9}{100}$ — 2 decimal places means hundredths $\left(\frac{}{100}\right)$

**b** $0.274 = \frac{274}{1000}$ — 3 decimal places means thousandths $\left(\frac{}{1000}\right)$

$= \frac{137}{500}$ — Simplify the fraction

**c** $1.45 = 1 + \frac{45}{100}$ — 2 decimal places means hundredths $\left(\frac{}{100}\right)$

$= 1\frac{9}{20}$ — Simplify the fraction

*Note*: The number of decimal places in the decimal gives the number of 0s in the denominator of the fraction.

## Decimal places

- tenths have 1 decimal place $\left(\frac{1}{10}\text{has 1 zero}\right)$
- hundredths have 2 decimal places $\left(\frac{1}{100}\text{has 2 zeros}\right)$
- thousandths have 3 decimal places $\left(\frac{1}{1000}\text{has 3 zeros}\right)$
- ten thousandths have 4 decimal places $\left(\frac{1}{10000}\text{has 4 zeros}\right)$

## Example 5

Convert each fraction to a decimal.

**a** $\frac{22}{100}$ **b** $3\frac{7}{10}$ **c** 5 thousandths

### Solution

**a** $\frac{22}{100} = 0.22$ — 2 zeros → 2 decimal places

**b** $3\frac{7}{10} = 3.7$ — 1 zero → 1 decimal place

**c** 5 thousandths $= \frac{5}{1000}$ — 3 zeros → 3 decimal places

$= 0.005$

Skillsheet Fractions and decimals

Puzzle Decimals squaresaw 1

7.02

**EXERCISE 7.02** ANSWERS ON P. 585

## Decimals and fractions U F C

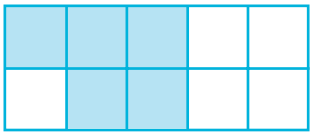
**1** What part of each shape has been shaded? Write the answer as: C

**i** a fraction **ii** a decimal

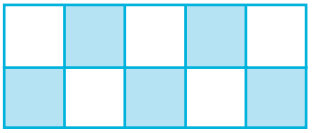
**a**



**b**



**c**



**d**



EXAMPLE 4

**2** Convert 0.028 to a simplified fraction. Select the correct answer **A**, **B**, **C** or **D**. C

**A** $\frac{28}{100}$ **B** $\frac{28}{1000}$ **C** $\frac{14}{500}$ **D** $\frac{7}{250}$

**3** Convert each decimal to a simplified fraction. C

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 0.7 | **b** | 0.4 | **c** | 0.39 | **d** | 0.572 |
| **e** | 0.003 | **f** | 0.05 | **g** | 0.11 | **h** | 0.328 |
| **i** | 0.3 | **j** | 0.999 | **k** | 0.013 | **l** | 0.0004 |
| **m** | 0.0471 | **n** | 0.321 | **o** | 0.5005 | **p** | 0.91 |
| **q** | 0.082 | **r** | 1.6 | **s** | 27.25 | **t** | 2.007 |

EXAMPLE 5

**4** Which decimal is equal to $\frac{23}{1000}$? Select **A**, **B**, **C** or **D**. C

**A** 0.23 **B** 0.023 **C** 0.0023 **D** 0.000 23

**5** Convert each fraction to a decimal. C

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $\frac{9}{10}$ | **b** | $\frac{15}{100}$ | **c** | $\frac{79}{100}$ | **d** | $\frac{60}{100}$ |
| **e** | 4 tenths | **f** | $\frac{23}{100}$ | **g** | $\frac{6}{10}$ | **h** | $\frac{411}{1000}$ |
| **i** | $\frac{704}{1000}$ | **j** | 87 hundredths | **k** | $\frac{7}{100}$ | **l** | $\frac{14}{100}$ |
| **m** | $\frac{8}{10}$ | **n** | $\frac{235}{1000}$ | **o** | $\frac{247}{1000}$ | **p** | $\frac{17}{100}$ |
| **q** | $\frac{368}{1000}$ | **r** | $\frac{9345}{10000}$ | **s** | $\frac{493}{1000}$ | **t** | $1\frac{67}{100}$ |
| **u** | $4\frac{47}{100}$ | **v** | $23\frac{9}{10}$ | **w** | $6\frac{8}{10}$ | **x** | 23 thousandths |
| **y** | 45 hundredths | | | | | | |

Foundation Standard Complex


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# Adding and subtracting decimals 7.03

## Adding and subtracting decimals

- Keep decimal points below one another
- Check your answer by estimating

Video
Adding and subtracting decimals



## Example 6

Find the sum of 10.92, 0.89, 32 and 0.6.

### Solution

$$\begin{array}{r} 10.92 \\ 0.89 \\ 32.00 \\ +\ 0.60 \\ \hline 44.41 \end{array}$$

Fill in any spaces with 0s.

Remember that 32 is the same as 32.00.

Estimate: $10.92 + 0.89 + 32 + 0.6 \approx 11 + 1 + 32 + 1$
$= 45$

## Example 7

Evaluate each difference.

**a** $76.029 - 8.914$ **b** $4.31 - 2.183$

### Solution

**a**
$$\begin{array}{r} 76.029 \\ -\ 8.914 \\ \hline 67.115 \end{array}$$

Estimate: $76.029 - 8.914 \approx 76 - 9$
$= 67$

**b**
$$\begin{array}{r} 4.310 \\ -\ 2.183 \\ \hline 2.127 \end{array}$$

Fill in any spaces with 0s.

Estimate: $4.31 - 2.183 \approx 4 - 2$
$= 2$

**EXERCISE 7.03** ANSWERS ON P. 585

## Adding and subtracting decimals U F PS R C

**1** Copy and complete each sum.

EXAMPLE 6

**a**
$$\begin{array}{r} 5.3 \\ 6.2 \\ +\ 0.5 \\ \hline \end{array}$$

**b**
$$\begin{array}{r} 4.723 \\ 0.01 \\ +\ 12.2 \\ \hline \end{array}$$

**c**
$$\begin{array}{r} 43.5 \\ 116.29 \\ 7.3 \\ +\ 0.227 \\ \hline \end{array}$$

**d**
$$\begin{array}{r} 0.0076 \\ 1.23 \\ +\ 0.9 \\ \hline \end{array}$$

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EXAMPLE 7

**2** Copy and complete each difference.

**a** $\begin{array}{r} 57.703 \\ -\ 16.21 \\ \hline \end{array}$ **b** $\begin{array}{r} 6.1 \\ -\ 0.2 \\ \hline \end{array}$ **c** $\begin{array}{r} 23.57 \\ -\ 16.88 \\ \hline \end{array}$ **d** $\begin{array}{r} 22.6 \\ -\ 13.54 \\ \hline \end{array}$

**3** Evaluate 0.61 + 12.345. Select the correct answer **A**, **B**, **C** or **D**.

**A** 73.345 **B** 18.445 **C** 12.955 **D** 12.406

**4** An electrician needed these lengths of cable to complete a wiring job: 12.3 m, 4.8 m, 18.7 m, 7.98 m, 13.65 m and 23.6 m. PS

**a** How many metres of cable did the electrician use?

**b** If the full spool of cable was 100 m long, how many metres of cable were left on the spool after the electrician completed the job?

**5** To keep fit, Angela runs each day. Last week she ran 3.8 km, 4.1 km, 2.3 km, 2.6 km, 3 km, 1 km and 1.8 km. How far did she run last week? PS

**6** A truck carrying sand had a total mass of 13 248 kg. If the truck alone had a mass of 5210.8 kg, what is the mass of the sand? PS

**7** 5 runners in the school's 100 m race recorded the following times: 13.5 s, 13.81 s, 12.7 s, 14.62 s, 12.45 s

**a** Place these times in order, from fastest to slowest.

**b** What is the time difference between the fastest and slowest runners?

**c** If the runner in second place had run 0.3 seconds faster, would she have won the race? Explain your answer. R C

**8** Dionne's expenses for one week are shown in the table.

**a** How much did he spend?

**b** How much did he have left out of his weekly pay of $1200.40?

| | |
|---|---|
| Food | $265.80 |
| Clothing | $88.45 |
| Car | $123.65 |
| Rent | $385.00 |
| Entertainment | $98.95 |
| Savings | $66.00 |

**9** A new block of wood is 11.27 cm thick. If it has 0.34 cm shaved off one side and 0.55 cm shaved off the other side, how thick is the block of wood? Select the correct answer **A**, **B**, **C** or **D**.

**A** 10.38 cm **B** 11.06 cm **C** 11.16 cm **D** 11.48 cm

**10** Wendy was making teddy bears. She needed these amounts of material for 5 bears: 2.6 m, 0.8 m, 1.2 m, 0.75 m and 0.88 m. How much material did she need altogether?

**11** Copy and complete each equation with the correct decimals. R

**a** 4.8 + _____ = 7.34 **b** _____ − 6.6 = 2.14

**c** 8.75 − _____ = 3.3 **d** _____ + _____ = 10.719

**e** _____ − _____ = 3.016

Foundation Standard Complex


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## Investigation

### Multiplying and dividing decimals by powers of 10

When a number is multiplied by 10, 100, 1000 or other powers of 10, every digit in the number increases in place value by moving to the left. When a number is divided by a power of 10, every digit decreases in place value by moving to the right.

**1** Copy the place value table below, then use a calculator to evaluate each expression involving multiplying or dividing by a power of 10. Note the pattern in your answers.

| Calculation | Ten thousands | Thousands | Hundreds | Tens | Units | tenths | hundredths | thousandths | ten thousandths |
|---|---|---|---|---|---|---|---|---|---|
| 36.7 | | | | 3 | 6 | 7 | | | |
| 36.7 × 10 | | | | | | | | | |
| 36.7 × 100 | | | | | | | | | |
| 36.7 × 1000 | | | | | | | | | |
| 2.35 | | | | | 2 | 3 | 5 | | |
| 2.35 × 10 | | | | | | | | | |
| 2.35 × 100 | | | | | | | | | |
| 2.35 × 1000 | | | | | | | | | |
| 36.7 ÷ 10 | | | | | | | | | |
| 36.7 ÷ 100 | | | | | | | | | |
| 36.7 ÷ 1000 | | | | | | | | | |
| 2.35 ÷ 10 | | | | | | | | | |
| 2.35 ÷ 100 | | | | | | | | | |

**2** Copy and complete each statement.

**a** To multiply by 10, move the decimal point _____ place to the _____.

**b** To multiply by 100, move the decimal point _____ places to the _____.

**c** To multiply by 1000, move the decimal point _____ places to the _____.

**d** To divide by 10, move the decimal point _____ place to the _____.

**e** To divide by 100, move the decimal point _____ places to the _____.

**f** To divide by 1000, move the decimal point _____ places to the _____.

# 7.04 Multiplying and dividing decimals by powers of 10

## Multiplying and dividing decimals by powers of 10

| To multiply a decimal by: | Move the decimal point: |
|---|---|
| 10 | 1 place to the right |
| 100 | 2 places to the right |
| 1000 | 3 places to the right |
| **To divide a decimal by:** | **Move the decimal point:** |
| 10 | 1 place to the left |
| 100 | 2 places to the left |
| 1000 | 3 places to the left |

Note that the number of 0s in the power of 10 is how many places the decimal point is moved.

**Video**
Multiplying and dividing decimals by powers of 10

## Example 8

Evaluate each expression.

**a** $13.69 \times 100$ **b** $2.54 \times 1000$ **c** $13.69 \div 10$ **d** $2.54 \div 100$

### Solution

**a** $13.69 \times 100 = 13.\overset{\curvearrowright}{6}\overset{\curvearrowright}{9}$
$= 1369$
Move the decimal point 2 places to the right
Moving right makes the decimal bigger

**b** $2.54 \times 1000 = 2.5\overset{\curvearrowright}{4}\overset{\curvearrowright}{0}$
$= 2540$
Move the decimal point 3 places to the right: add a 0 to the end to allow this.

**c** $13.69 \div 10 = 1\overset{\curvearrowleft}{3}.69$
$= 1.369$
Move the decimal point 1 place to the left
Moving left makes the decimal smaller

**d** $2.54 \div 100 = 0\overset{\curvearrowleft}{0}\overset{\curvearrowleft}{2}.54$
$= 0.0254$
Move the decimal point 2 places to the left: add 2 0s to the start to allow this

**EXERCISE 7.04** ANSWERS ON P. 585

## Multiplying and dividing decimals by powers of 10 UFR

EXAMPLE 8

**1** Evaluate each product.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $2.49 \times 100$ | **b** | $0.81 \times 10$ | **c** | $37.42 \times 1000$ |
| **d** | $3.416 \times 100$ | **e** | $7.25 \times 100$ | **f** | $2.196 \times 1000$ |
| **g** | $6.043 \times 100$ | **h** | $0.032 \times 100$ | **i** | $0.065 \times 10$ |
| **j** | $45.213 \times 100$ | **k** | $10.64 \times 1000$ | **l** | $63.04 \times 100$ |
| **m** | $5.98 \times 1000$ | **n** | $847.612 \times 100$ | **o** | $0.0592 \times 10$ |
| **p** | $36.2 \times 100$ | **q** | $219.4 \times 1000$ | **r** | $40.075 \times 10$ |

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**2** Evaluate 18.5 ÷ 1000. Select the correct answer **A**, **B**, **C** or **D**.

**A** 0.00 185 **B** 1.85 **C** 0.0185 **D** 0.185

**3** Evaluate each quotient.

| | | | | | |
|---|---|---|---|---|---|
| **a** | 46.3 ÷ 10 | **b** | 507 ÷ 100 | **c** | 1203 ÷ 1000 |
| **d** | 36.4 ÷ 100 | **e** | 381.5 ÷ 1000 | **f** | 55.02 ÷ 10 |
| **g** | 11.46 ÷ 1000 | **h** | 694.3 ÷ 1000 | **i** | 4.28 ÷ 1000 |
| **j** | 67 ÷ 100 | **k** | 21.31 ÷ 1000 | **l** | 5.72 ÷ 10 |
| **m** | 81.348 ÷ 1000 | **n** | 50.2 ÷ 100 | **o** | 4.91 ÷ 100 |
| **p** | 0.4 ÷ 1000 | **q** | 1.73 ÷ 100 | **r** | 125.3 ÷ 10 000 |

**4** Copy and complete each equation. R

| | | | |
|---|---|---|---|
| **a** | 14.219 × ____ = 14 219 | **b** | 892.5 ÷ ____ = 8.925 |
| **c** | 742 ÷ ____ = 7.42 | **d** | 205.7 ÷ ____ = 20.57 |
| **e** | ____ × ____ = 560.1 | **f** | ____ ÷ ____ = 0.382 |

7.04

Quiz
Mental skills 7

## Mental skills 7: Maths without calculators ANSWERS ON P. 585

### Calculating differences and making change

In every subtraction problem, for example 135 – 47, think of finding the 'gap' between the 2 numbers. That is, find the number in this case that must be added to 47 to get 135.

**1** Study each example.

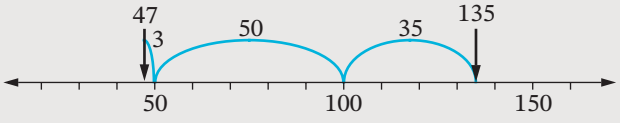
**a** 135 – 47

Think: 47 + ___ = 135

Count: '47, 50, 100, 135'

Add: 3 + 50 + 35 = 88

Answer: 135 – 47 = 88



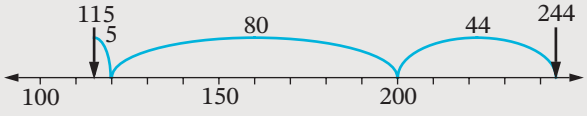

**b** 244 – 115

Think: 115 + ___ = 244

Count: '115, 120, 200, 244'

Add: 5 + 80 + 44 = 129

Answer: 244 – 115 = 129

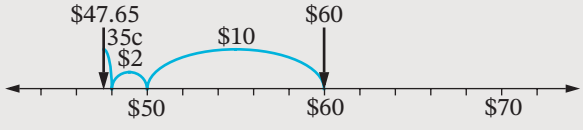



**c** $60 – $47.65

Count: '$47.65, $48, $50, $60'

Add: $0.35 + $2 + $10 = $12.35

Answer: $60 – $47.65 = $12.35



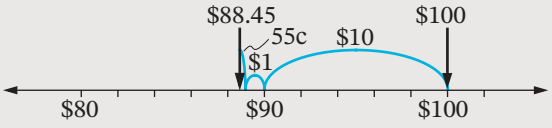

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**d** $100 – $88.45

Count: '$88.45, $89, $90, $100'

Add: $0.55 + $1 + $10 = $11.55

Answer: $100 – $88.45 = $11.55




**2** Now evaluate each difference.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 176 – 88 | **b** | 221 – 54 | **c** | 670 – 356 | **d** | 425 – 340 |
| **e** | 518 – 389 | **f** | 199 – 78 | **g** | $70 – $58.40 | **h** | $80 – $73.25 |
| **i** | $45 – $40.30 | **j** | $100 – $69.95 | **k** | $30 – $22.90 | **l** | $50 – $17.10 |

# 7.05 Multiplying decimals by estimating

If you know the answer to a whole number multiplication, then you can use estimation to find the answer to a related decimal multiplication by correctly positioning the decimal point.

## Example 9

**a** Given that $17 \times 12 = 204$, find:

**i** $1.7 \times 12$ **ii** $1.7 \times 1.2$

**b** Given that $23 \times 47 = 1081$, find:

**i** $2.3 \times 4.7$ **ii** $230 \times 4.7$ **iii** $23 \times 0.47$

### Solution

**a** $17 \times 12 = 204$

| Multiplication | Estimate | Answer | |
|---|---|---|---|
| $1.7 \times 12$ | $\approx 2 \times 10 = 20$ | 20.4 | ← use the digits 204 to make a number near 20 |
| $1.7 \times 1.2$ | $\approx 2 \times 1 = 2$ | 2.04 | ← use the digits 204 to make a number near 2 |

**b** $23 \times 47 = 1081$

| Multiplication | Estimate | Answer | |
|---|---|---|---|
| $2.3 \times 4.7$ | $\approx 2 \times 5 = 10$ | 10.81 | ← use the digits 1081 to make a number near 10 |
| $230 \times 4.7$ | $\approx 200 \times 5 = 1000$ | 1081 | ← use the digits 1081 to make a number near 1000 |
| $23 \times 0.47$ | $\approx 20 \times 0.5 = 10$ | 10.81 | ← use the digits 1081 to make a number near 10 |

**EXERCISE 7.05** ANSWERS ON P. 586

## Multiplying decimals by estimating U F

EXAMPLE 9

7.05

**1** Use the result of the whole number multiplication to copy and complete each table.

**a** $69 \times 18 = 1242$

| Multiplication | Estimate | Answer |
|---|---|---|
| 0.69 × 18 | | |
| 6.9 × 180 | | |
| 6.9 × 1.8 | | |
| 690 × 1.8 | | |

**b** $104 \times 42 = 4368$

| Multiplication | Estimate | Answer |
|---|---|---|
| 10.4 × 42 | | |
| 1.04 × 4.2 | | |
| 104 × 4.2 | | |
| 0.104 × 4.2 | | |

**c** $38 \times 92 = 3496$

| Multiplication | Estimate | Answer |
|---|---|---|
| 3.8 × 92 | | |
| 0.38 × 92 | | |
| 38 × 0.92 | | |
| 380 × 0.92 | | |

**2** Given that $63 \times 34 = 2142$, use estimates to find:

**a** $6.3 \times 3.4$ **b** $0.63 \times 3.4$ **c** $0.63 \times 3400$

**d** $630 \times 3.4$ **e** $6.3 \times 34$ **f** $63 \times 0.34$

**3** Given that $1.7 \times 1.2 = 2.04$, use estimates to find:

**a** $1.7 \times 12$ **b** $17 \times 0.12$ **c** $0.17 \times 1.2$

**d** $0.17 \times 12$ **e** $17 \times 12$ **f** $17 \times 1.2$

**4** Given that $7.2 \times 3.4 = 24.48$, use estimates to find:

**a** $7.2 \times 34$ **b** $72 \times 3.4$ **c** $0.72 \times 3.4$

**d** $72 \times 34$ **e** $7.2 \times 0.34$ **f** $0.72 \times 34$

**5** Given that $1.26 \times 6 = 7.56$, use estimates to find:

**a** $12.6 \times 6$ **b** $126 \times 6$ **c** $1.26 \times 0.6$

**d** $0.126 \times 6$ **e** $0.126 \times 0.6$ **f** $126 \times 0.6$

**6** Use the fact that $0.3 \times 0.24 = 0.072$ to find:

**a** $3 \times 0.24$ **b** $0.3 \times 2.4$ **c** $0.3 \times 24$

**d** $3 \times 24$ **e** $3 \times 2.4$ **f** $0.03 \times 24$

**7** Write an estimate for each product.

**a** $2.7 \times 3.1$ **b** $6.2 \times 0.44$ **c** $8.6 \times 19$

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## Investigation

### Decimal places in multiplication answers

**1** What happens when you multiply by a number less than one?

- **a** Is the answer to $12 \times 0.8$ more or less than 12? Why?
- **b** Estimate the answer to $12 \times 0.8$.
- **c** How many decimal places do 12 and 0.8 each have?
- **d** Use a calculator to evaluate $12 \times 0.8$. How many decimal places does the answer have?

**2**
- **a** Is the answer to $0.7 \times 0.3$ more or less than 0.7? Why?
- **b** Estimate the answer to $0.7 \times 0.3$.
- **c** How many decimal places do 0.7 and 0.3 each have?
- **d** Use a calculator to evaluate $0.7 \times 0.3$. How many decimal places does the answer have?

**3**
- **a** Is the answer to $2.5 \times 4.1$ more or less than 2.5? Why?
- **b** Estimate the answer to $2.5 \times 4.1$.
- **c** How many decimal places do 2.5 and 4.1 each have?
- **d** Use a calculator to evaluate $2.5 \times 4.1$. How many decimal places does the answer have?

**4** What is the relationship between the number of decimal places in the question and the number of decimal places in the answer?

**5**
- **a** If $82 \times 6 = 492$, what do you think is the answer to $82 \times 0.6$? Where does the decimal point go?
- **b** If $4 \times 17 = 68$, what do you think is the answer to $0.4 \times 1.7$?
- **c** If $367 \times 51 = 18\,717$, what do you think is the answer to $3.67 \times 5.1$?
- **d** What is the answer to $0.5 \times 0.9$?

# Multiplying decimals 7.06

*Note:* Multiplying and dividing decimals (including by whole numbers) using written methods is no longer in the syllabus. Teachers can use judgement to teach calculator methods only.

## Multiplying decimals

When **multiplying decimals**, the number of decimal places in the answer equals the total number of decimal places in the question.

## Example 10

Evaluate each product.

**a** $0.6 \times 4.1$ **b** $3.06 \times 4.8$

### Solution

**a** Complete the multiplication without decimal points.

$$\begin{array}{r} 41 \\ \times\ 6 \\ \hline 246 \end{array}$$

Decide where to place the decimal point by counting decimal places or by estimating.

**Method 1**

0.6 has 1 decimal place, 4.1 has 1 decimal place.

So the answer has 2 decimal places.

Insert the decimal point so that the answer has 2 decimal places: $0.6 \times 4.1 = \mathbf{2.46}$

**Method 2**

Estimate the answer: $0.6 \times 4.1 \approx 1 \times 4 = 4$

Insert the decimal point so that the answer is near 4: $0.6 \times 4.1 = \mathbf{2.46}$

**b** Complete the multiplication without decimal points.

$$\begin{array}{r} 306 \\ \times\ 48 \\ \hline 2448 \\ 12240 \\ \hline 14688 \end{array}$$

**Method 1**

3.06 has 2 decimal places, 4.8 has 1 decimal place.

So we need $2 + 1 = 3$ decimal places.

Insert the decimal point so that the answer has 3 decimal places: $3.06 \times 4.8 = \mathbf{14.688}$



**Worksheets**
Shopping and change
Decimal cards
Where's the point?

**Video**
Multiplying decimals

**Puzzles**
Which decimals?
Multiplication estimation game

**Interactive**
Area Model Decimals

**Method 2**

Estimate the answer: $3.06 \times 4.8 \approx 3 \times 5 = 15$

Insert the decimal point so that the answer is near 15.

$3.06 \times 4.8 =$ **14.688**

## EXERCISE 7.06 ANSWERS ON P. 586

## Multiplying decimals U F PS C

**1** How many decimal places will each product have? C

| | | | | | |
|---|---|---|---|---|---|
| **a** | $0.25 \times 11$ | **b** | $10.2 \times 4$ | **c** | $0.5 \times 10$ |
| **d** | $7 \times 2.193$ | **e** | $0.9 \times 0.75$ | **f** | $8.06 \times 4.1$ |
| **g** | $0.11 \times 1.01$ | **h** | $6.3 \times 0.04$ | **i** | $2.95 \times 5.3$ |
| **j** | $0.237 \times 1.2$ | **k** | $0.023 \times 0.042$ | **l** | $321.2 \times 8.1$ |

EXAMPLE 10

**2** Evaluate $0.4 \times 5.76$. Select the correct answer **A**, **B**, **C** or **D**.

**A** 0.2304 **B** 2.304 **C** 23.04 **D** 230.4

**3** Evaluate each product.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | $3.05 \times 4$ | **b** | $1.02 \times 7$ | **c** | $2.001 \times 9$ | **d** | $17.1 \times 2$ |
| **e** | $10 \times 2.25$ | **f** | $3 \times 4.2$ | **g** | $6.95 \times 5$ | **h** | $1.004 \times 8$ |
| **i** | $0.18 \times 5$ | **j** | $0.4 \times 12$ | **k** | $6 \times 0.002$ | **l** | $8 \times 6.75$ |
| **m** | $0.4 \times 0.8$ | **n** | $3.9 \times 0.5$ | **o** | $0.8 \times 0.6$ | **p** | $0.3 \times 0.24$ |

**4** Estimate each product, then evaluate the product.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $47.9 \times 0.3$ | **b** | $6.43 \times 7.2$ | **c** | $83.4 \times 6.3$ |

**5** Dress material costs $31.24 per metre. How much does 5.2 metres of material cost?

**6** Sean earns $28.45 per hour working at a cinema. How much will he earn for working a 39-hour week?

**7** A flower bed is 2.6 m long by 0.8 m wide. Calculate its area.

**8** Calculate the total of this grocery shopping bill. Select **A**, **B**, **C** or **D**. PS

- 500 g of butter at $1.20 per 100 g
- 500 g of cheese at $8.40 per kilogram
- 2 kg of chicken at $9.50 per kilogram
- 2 dozen eggs at $5.95 per dozen.

**A** $25.05 **B** $35.70 **C** $41.10 **D** $53.70



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# Dividing decimals by whole numbers 7.07

*Note:* Multiplying and dividing decimals (including by whole numbers) using written methods is no longer in the syllabus. Teachers can use judgement to teach calculator methods only.

## Dividing decimals by whole numbers

**When dividing a decimal by a whole number:**

- rewrite the question in 'short division' form
- make the decimal point in the answer line up with the decimal point in the question
- add 0s to the end of the decimal being divided, if needed

## Example 11

Evaluate each quotient.

**a** $10 \div 4$ **b** $0.018 \div 6$ **c** $2.66 \div 8$

### Solution

**a**

$$4\overline{)10.^{2}0}\quad 2.\ 5$$

$10 \div 4 = 2.5$

Write 10 as 10.0 so that you can complete the division

**b**

$$6\overline{)0.01^{1}8}\quad 0.00\ 3$$

$0.018 \div 6 = 0.003$

**c**

$$8\overline{)2.6^{2}6^{2}0^{4}0}\quad 0.3\ 3\ 2\ 5$$

$2.66 \div 8 = 0.3325$

Write 2 zeros after 2.66 so that you can complete the division

**EXERCISE 7.07** ANSWERS ON P. 586

## Dividing decimals by whole numbers U F R

EXAMPLE 11

**1** Evaluate each quotient.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $4.8 \div 2$ | **b** | $18.6 \div 3$ | **c** | $20.8 \div 5$ |
| **d** | $32.8 \div 8$ | **e** | $29.3 \div 2$ | **f** | $8.79 \div 4$ |
| **g** | $0.056 \div 7$ | **h** | $10.71 \div 4$ | **i** | $195.6 \div 8$ |
| **j** | $7.35 \div 2$ | **k** | $4.15 \div 8$ | **l** | $0.318 \div 3$ |
| **m** | $12 \div 5$ | **n** | $13.56 \div 4$ | **o** | $23 \div 8$ |
| **p** | $256.84 \div 4$ | **q** | $107.1 \div 9$ | **r** | $82.5 \div 6$ |
| **s** | $0.732 \div 6$ | **t** | $2075.6 \div 8$ | | |

**2** Yesterday, Anh earned $225.92 for working 8 hours at the chemist. How much was he paid per hour?

**3** Trent took 5 hours to travel 163.5 km on his bike. What was his average speed in km/h? Select the correct answer **A**, **B**, **C** or **D**.

**A** 30.6 **B** 32.6 **C** 8.175 **D** 32.7

**4** At the restaurant, the bill for 6 friends totalled $366.96. How much should each friend pay if the bill is to be split evenly?

**5** Nadine ran 100 metres in the following times (in seconds):

11.4 12.1 11.5 11.3 11.6 11.7

**a** Calculate the sum of Nadine's 6 times.

**b** Calculate Nadine's average time.

**6** Copy and complete each equation with decimals. R

**a** _____ $\div 8 = 3.1$ **b** _____ $\div 5 = 2.46$ **c** _____ $\div$ _____ $= 6$

# 7.08 Dividing decimals

Look at this pattern:

$18 \div 3 = 6$

$180 \div 30 = 6$

$1800 \div 300 = 6$

**Worksheets**
Decimals writing activity
Decimals 7
Decimal number grids

This is like writing them as equivalent fractions: $\frac{18}{3} = \frac{180}{30} = \frac{1800}{300}$.

When dividing numbers, if we multiply both numbers by the **same number** first, the answer stays the same. We can use this property to help us divide decimals. For example:

$9.8 \div 0.08 = 980 \div 8$ (multiplying both numbers by 100)

$= 122.5$

**Puzzle**
Operations with decimals

$980 \div 8$ is easier to evaluate than $9.8 \div 0.08$ because 8 is a whole number.

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## Dividing decimals

**To divide a decimal by a decimal:**

- make the second decimal a whole number by moving the decimal point the required number of places to the right
- move the point in the first decimal the same number of places to the right
- divide the new first number by the whole number

This works because we multiply both decimals by the same power of 10 before dividing.



Video
Dividing decimals

## Example 12

Evaluate each quotient:

**a** $0.4 \div 0.2$ **b** $1.75 \div 0.5$ **c** $122.4 \div 0.03$

### Solution

**a** $0.4 \div 0.2 = 0.4 \div 0.2$
$= 4 \div 2$
$= 2$

multiplying both decimals by 10

Move both decimal points one place to the right so that 0.2 becomes a whole number

**b** $1.75 \div 0.5 = 1.75 \div 0.5$
$= 17.5 \div 5$
$= 3.5$

Move both decimal points one place to the right

**c** $122.4 \div 0.03 = 122.40 \div 0.03$
$= 12\,240 \div 3$
$= 4080$

multiplying both decimals by 100

Move both decimal points 2 places to the right so that 0.03 becomes a whole number

**EXERCISE 7.08** ANSWERS ON P. 586

## Dividing decimals U F R C

**1** Rewrite each division expression so that the second decimal is a whole number.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $508.8 \div 0.8$ | **b** | $17.92 \div 0.07$ | **c** | $333 \div 0.9$ |
| **d** | $1.725 \div 0.05$ | **e** | $129.2 \div 0.4$ | **f** | $49.5 \div 0.06$ |
| **g** | $168 \div 0.7$ | **h** | $14.823 \div 0.09$ | **i** | $0.66 \div 0.3$ |

**2** Evaluate each quotient, and check that your answers seem reasonable by estimating.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $3.48 \div 0.4$ | **b** | $7.32 \div 0.2$ | **c** | $2.94 \div 0.6$ |
| **d** | $16.28 \div 0.08$ | **e** | $27 \div 0.9$ | **f** | $10.08 \div 0.8$ |
| **g** | $10.4 \div 0.05$ | **h** | $5.6 \div 0.07$ | **i** | $1.71 \div 0.3$ |
| **j** | $40.82 \div 0.02$ | **k** | $0.532 \div 0.5$ | **l** | $0.7812 \div 0.006$ |

EXAMPLE 12

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**3** **a** $18 \div 0.5$ means 'how many times does 0.5 go into 18?'.
Is the answer more or less than 18? Why?

**b** Estimate the answer to $18 \div 0.5$.

**c** Evaluate $18 \div 0.5$.

**d** Show that $18 \div \frac{1}{2}$ gives the same answer. R C

**4** **a** $20.4 \div 0.3$ means 'how many times does 0.3 go into 20.4?'.
Is the answer more or less than 20.4?

**b** Estimate the answer to $20.4 \div 0.3$.

**c** Find the exact answer to $20.4 \div 0.3$. R C

**5** What happens when you divide by a number less than 1? Is the answer more or less than the number? (Check your answers to question **2**). R C

**6** Which of the following is the answer to $13.59 \div 0.03$? Select **A**, **B**, **C** or **D**.

**A** 45.3 **B** 453 **C** 4.53 **D** 4530

**7** A drink bottle holds 0.8 litres. How many drink bottles can be filled from a container that holds 12.8 litres?

**8** A square wall tile has length 0.3 m. If a bathroom wall has length 5.2 m, how many whole tiles are needed for one row along the wall? R

**9** James can cycle 0.7 km in 1 minute.

**a** How far will he travel in 20 minutes?

**b** How long will it take him to travel 26.6 km?

**10** Copy and complete each equation with the correct decimals. R

**a** $27.2 \div$ _____ $= 8$ **b** _____ $\div 5 = 2.46$ **c** _____ $\div$ _____ $= 6.2$

## Investigation

### Back-to-front problems

The cards for this set of questions have been printed without any decimal points. Insert the decimal points so that the numbers on the cards fit the clues.

**1** The difference between these 2 numbers is 53.7. The sum of the numbers is 58.5.

| 561 | 24 |
|---|---|

**2** The difference between the numbers is 178.8. When you divide the greater number by the smaller number, the quotient is between 43 and 44.

| 183 | 42 |
|---|---|

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**3** The sum of the 3 numbers is 5.36.The product of the numbers is 1.2096.

8 | 42 | 36

**4** The sum of the 3 numbers is 4.61. The product of the numbers is 0.9135.

15 | 29 | 21

**5** The sum of the 4 numbers is 2.55.The product of 2 of the numbers is 0.196. The product of the other 2 numbers is 0. 217.

7 | 31 | 14 | 14



# Terminating and recurring decimals 7.09

When fractions are converted to decimals, the decimals can be **terminating** or **recurring**. **Terminating decimals** have a definite number of decimal places, such as 0.625, while **recurring decimals** are decimals whose digits repeat endlessly, such as 0.272727... 'Terminate' means 'to stop' while 'recurring' means 'repeating'.

**Skillsheet** Fractions and decimals

**Puzzle** Decimals squaresaw 2

**Video** Converting fractions to decimals

## Terminating decimals

### Example 13

Convert each fraction into a decimal.

**a** $\frac{3}{5}$ **b** $\frac{5}{8}$

### Solution

**a** $\frac{3}{5}$ means $3 \div 5$

$$5\overline{)3.0}\quad 0.6$$

**b** $\frac{5}{8}$ means $5 \div 8$

$$8\overline{)5.000}\quad 0.625$$

(2, 4)

Remember to add 0s, if necessary, to complete the division.

$\frac{3}{5} = 0.6$

$\frac{5}{8} = 0.625$

Note also that $\frac{3}{5} = \frac{6}{10} = 0.6$.

# Recurring decimals

**Videos**
Converting fractions to decimals
Recurring decimals

## Example 14

Convert each fraction to a decimal.

**a** $\frac{1}{3}$ **b** $\frac{5}{6}$ **c** $\frac{2}{11}$

### Solution

**a** $\frac{1}{3}$ means $1 \div 3$

$$3\overline{)1.000\ldots}\quad 0.333\ldots$$

$\frac{1}{3} = 0.333... = 0.\dot{3}$ or $0.\overline{3}$

**b** $\frac{5}{6}$ means $5 \div 6$

$$6\overline{)5.0000\ldots}\quad 0.833\ldots$$

$\frac{5}{6} = 0.8333... = 0.8\dot{3}$ or $0.8\overline{3}$

**c** $\frac{2}{11}$ means $2 \div 11$

$$11\overline{)2.00000\ldots}\quad 0.18181\ldots$$

$\frac{2}{11} = 0.181\,818... = 0.\dot{1}\dot{8}$ or $0.\overline{18}$

The answers in the previous example are all **recurring decimals**, where one or more of the digits in the decimal repeat endlessly. To show this, we use dots or a line to mark the repeating section: for example, $0.259\,259\,259... = 0.2\dot{5}\dot{9}$ or $0.\overline{259}$.

**EXERCISE 7.09** ANSWERS ON P. 586

## Terminating and recurring decimals U F R C

EXAMPLE 13

**1** Copy and complete this table. Use a calculator if you need to.

| Common fraction | Meaning as division | Decimal |
|---|---|---|
| $\frac{3}{5}$ | $3 \div 5$ | 0.6 |
| $\frac{1}{2}$ | $1 \div 2$ | |
| $\frac{1}{4}$ | | 0.25 |
| $\frac{4}{5}$ | | |
| $\frac{2}{5}$ | | |
| | $3 \div 4$ | 0.75 |
| | $1 \div 5$ | |
| | $1 \div 8$ | |

**2** Use your answers from question **1** to help you convert each fraction into a decimal.

**a** $\frac{2}{5}$ **b** $\frac{3}{8}$ **c** $\frac{3}{4}$ **d** $\frac{2}{2}$

**e** $\frac{2}{4}$ **f** $\frac{6}{8}$ **g** $\frac{3}{5}$ **h** $\frac{2}{8}$

**i** $\frac{5}{8}$ **j** $\frac{7}{8}$ **k** $\frac{4}{8}$ **l** $\frac{5}{5}$

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**3** Explain why some of the fractions in question **2** have the same decimal value. R C

**4** Write each mixed numeral as a decimal.

**a** $4\frac{8}{10}$ **b** $23\frac{3}{4}$ **c** $12\frac{5}{8}$ **d** $6\frac{3}{5}$

**e** $57\frac{2}{5}$ **f** $19\frac{1}{8}$ **g** $110\frac{7}{8}$ **h** $80\frac{1}{4}$

**5** Rewrite each recurring decimal using the dot notation. C

**a** 0.41666666 ... **b** 0.27272727...

**c** 0.111111 ... **d** 0.1027510275 ...

**6** Write each recurring decimal showing the repeated pattern. C

**a** $0.858\dot{3}$ **b** $0.0\dot{4}\dot{5}$ **c** $0.\dot{7}$ **d** $0.\dot{4}61538\dot{8}$

**7** Convert each fraction to a recurring decimal. C

EXAMPLE 14

**a** $\frac{1}{9}$ **b** $\frac{1}{6}$ **c** $\frac{5}{6}$ **d** $\frac{1}{7}$ **e** $\frac{2}{3}$

**f** $\frac{2}{7}$ **g** $\frac{2}{9}$ **h** $\frac{3}{7}$ **i** $\frac{4}{7}$ **j** $\frac{4}{9}$

**k** $\frac{4}{6}$ **l** $\frac{5}{9}$ **m** $\frac{6}{7}$ **n** $\frac{7}{9}$ **o** $\frac{8}{9}$

**8** Copy and complete the table and note the pattern. R C

| Fraction | $\frac{1}{9}$ | $\frac{2}{9}$ | $\frac{3}{9}$ | $\frac{4}{9}$ | $\frac{5}{9}$ | $\frac{6}{9}$ | $\frac{7}{9}$ | $\frac{8}{9}$ |
|---|---|---|---|---|---|---|---|---|
| Decimal | | | | | | | | |

7.10

# Rounding decimals 7.10

Sometimes, to approximate an answer with many decimal places, we round to fewer decimal places. We need to be able to **round** when working with money, measuring quantities or writing answers to division calculations.

## Rounding decimals

To round a decimal, 'cut' it at the required decimal place and look at the digit in the next place:

- if the digit is less than 5 (that is, 0, 1, 2, 3 or 4), **round down**
- if the digit is 5 or more (that is, 5, 6, 7, 8 or 9), **round up**

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Video
Rounding decimals

## Example 15

**a** Round 86.246 to:

**i** one decimal place **ii** 2 decimal places

**b** Write 0.087 1245 correct to:

**i** one decimal place **ii** the nearest thousandth

### Solution

**a i** 86.2|46

cut the next digit is 4 (less than 5), so round **down**

So 86.246 ≈ 86.2 (correct to one decimal place).

**ii** 86.24|6

cut the next digit is 6 (more than 5), so round **up** to 86.25

So 86.246 ≈ 86.25 (correct to 2 decimal places).

**b i** 0.0|87 1245

cut the next digit is 8 (more than 5), so round **up** to 0.1

0.087 1245 ≈ 0.1

**ii** 0.087|1245

cut the next digit is 1 (less than 5), so round **down**

'the nearest thousandth' means 3 decimal places

0.087 1245 ≈ 0.087

## Rounding to decimal places

Rounding to:

- the nearest tenth = 1 decimal place
- the nearest hundredth = 2 decimal places
- the nearest thousandth = 3 decimal places

**EXERCISE 7.10** ANSWERS ON P. 587

## Rounding decimals UFRC

EXAMPLE 15

**1** Write each decimal correct to one decimal place. C

| | | | |
|---|---|---|---|
| **a** 0.35 | **b** 0.47 | **c** 0.81 | **d** 0.69 |
| **e** 2.55 | **f** 0.32 | **g** 0.90 | **h** 2.88 |

**2** Round each decimal to 2 decimal places. C

| | | | |
|---|---|---|---|
| **a** 0.481 | **b** 0.736 | **c** 0.069 | **d** 0.293 |
| **e** 0.309 | **f** 0.655 | **g** 2.096 | **h** 3.995 |

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**3** Copy and complete this table. Use a calculator to help you. C

| | Question | Calculator display | Rounded to the nearest tenth | Rounded to the nearest hundredth |
|---|---|---|---|---|
| **a** | 12.19 ÷ 3 | 4.0633333333 | 4.1 | 4.06 |
| **b** | 12.32 ÷ 6 | | | |
| **c** | 19.82 ÷ 9 | | | |
| **d** | 56.85 ÷ 11 | | | |
| **e** | 17.13 ÷ 4 | | | |
| **f** | 12.65 ÷ 12 | | | |
| **g** | 4.875 ÷ 21 | | | |
| **h** | 27.45 ÷ 8 | | | |
| **i** | 17 ÷ 12 | | | |
| **j** | 254.678 ÷ 32 | | | |

**4** Write each decimal correct to 2 decimal places. C

**a** 25.3759 **b** 341.6143 **c** 420.8971

**d** 734.6541 **e** 27.359 94 **f** 1314.210 47

**5** Round each number to 4 decimal places. C

**a** 10.333 74 **b** 431.543 27 **c** 1.444 95

**d** 3217.654 061 **e** 4.670 89 **f** 0.888 88

**6** The calculated answers to the following problems are decimals but, for particular reasons, some need to be rounded up and some need to be rounded down. Find each rounded answer, giving the reason why you rounded up or down. R C

**a** A box of chocolates with 44 chocolates is shared among a family of 5 people. How many chocolates does each person receive?

**b** A new bathroom requires 32 square metres of tiles. A box of tiles covers 0.6 square metres of area. How many boxes are needed to tile the bathroom?

**c** A team of 4 golfers wins 27 new golf balls in a competition. How many does each person receive?

**d** Some timber comes in 0.9 m lengths. How many lengths are needed to build a chicken house needing 23 m of timber?

**e** One dress requires 1.3 m of material. How many dresses can be made from a 5 m length of material?

**7** Write a decimal that could be rounded to: R

**a** 2.718 **b** 0.8 **c** 25.6

**8** Australia had 1c coins and 2c coins until 1990. After that, supermarket prices had to be rounded to the nearest 5 cents. Find out how prices are rounded this way. R C

## Technology

### Rounding decimals

This activity will show you how to format a cell so that its value is rounded to a specific number of decimal places. Start a new spreadsheet.

1. Enter **748.61** into cell A1, **x** (for multiply) into cell A2 and **3.75** into cell A3.
2. Enter a formula into cell A4 for the product of A1 and A3 and **bold** your answer.
3. To round this answer to 1 decimal place, **right-click** on cell A4, choose **Format Cells**, **Number**, and 1 decimal place.
4. Enter **748.61** into cell B1, '**/** (for divide, including the ') into cell B2 and **3.75** into cell B3.
5. Enter a formula into cell B4 for B1 divided by B3 and **bold** your answer.
6. To round this answer to 2 decimal places, **right-click** on cell B4, choose **Format Cells**, **Number**, and 2 decimal places.
7. Complete each calculation using your spreadsheet and creating appropriate formulas.
   - **a** 284.796 ÷ 32.4
   - **b** 1217.9 ÷ 45.6 (round to 3 decimal places)
   - **c** 1.15 ÷ 1.5 (round to 2 decimal places)
   - **d** 1604.12 ÷ 0.02 ÷ 4.578
   - **e** (8756.32 − 9025.198 + 1023.5697) ÷ 1.444 (round to 4 decimal places)
   - **f** (12.3 + 6.59) ÷ (56.4 × 0.04) (round to 5 decimal places)

## Did you know?

### Scoring in Olympics gymnastics

How is the score for a gymnast in an Olympic competition calculated? Each gymnast starts with a score of 10.0 plus an allowance for the level of difficulty of the routine. Deductions are made from 10.0 for errors, missteps or falls. The deductions can be as low as 0.01 or multiples of 0.01 for small errors. Higher deductions for a fall or a bigger mistake can be 0.1 or multiples of 0.1.



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High-level competitions have 6 judges and each judge gives a score. The highest and lowest scores are removed, and the remaining 4 scores are averaged to give the athlete's score. The difficulty score is then added. Team events combine the scores of individual athletes.

**What were the highest scores in the last Olympic gymnastics competition?**

**In which events are the highest scores found?**

# Decimal problems 7.11

**EXERCISE 7.11** ANSWERS ON P. 587

## Decimal problems U F PS R C

**1** A farmer wants to fence a rectangular paddock. The paddock is 35.6 metres long and 20.85 metres wide. How many metres of fencing will be needed? PS

**2** A drink bottle holds 0.6 litres. How many drink bottles can be filled from a tub that holds 4.5 litres? PS

**3** A car travels 110.3 kilometres on 7 litres of petrol. How many kilometres would the car travel on one litre of petrol? (Give the answer correct to one decimal place.)

**4** Anja runs 3.8 kilometres each day of the week. How far does she run in one week?

**5** These calculator displays show amounts of money in decimal form. Rewrite each amount in dollars and cents, to the nearest cent. C

a 15.236 b 6.8412 c 487.759318 d 1234.047126

**6** Find the cost of 352 units of electricity at 25.3 cents per unit.
Select the correct answer **A**, **B**, **C** or **D**.

**A** $8905.60 **B** $890.56 **C** $89.06 **D** $8.91

**7** Mark buys golf balls for $4.85 each and sells them for $5.15 each. How much money does he make if he sells 30 golf balls? PS

**8** A long distance train is made up of a diesel engine, 2 dining cars and 15 passenger carriages. The engine has a mass of 20.2 tonnes, each dining car has a mass of 14.35 tonnes and each passenger carriage has a mass of 13.96 tonnes. How heavy is the entire train? PS



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**9** Samir is cutting shelves from a board that is 4.6 metres long. Each shelf needs to be 0.9 metres long. How many shelves can be cut?

**Worksheets**
Shopping and change
Calculating change

7.11

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**10** The table lists some Sydney FM radio stations and their allocated frequencies measured in megahertz (MHz). R C

| Station | Frequency (MHz) | Station | Frequency (MHz) |
|---|---|---|---|
| C91.3 | 91.3 | Hope | 103.2 |
| ABC Classic | 92.9 | 2Day FM | 104.1 |
| Smooth FM | 95.3 | Triple M | 104.9 |
| The Edge | 96.1 | Triple J | 105.7 |
| Nova | 96.9 | KIIS | 106.5 |
| SBS | 97.7 | 2SER | 107.3 |
| WS-FM | 101.7 | | |

**a** Copy this number line, then locate the stations on it according to their frequencies.




**b** What is the frequency difference in megahertz between 2Day FM and WS-FM?

**c** Find the smallest frequency difference between adjacent stations. ('Adjacent' means side-by-side.)

**d** What is the largest difference in frequency between adjacent stations?

**11** A sheet of cardboard is 0.3 mm thick. How many sheets would be in a stack that is 3.6 cm high? PS

**12** Holly walks to work and back each day. She works 6 days a week and, in one week, walks 16.8 kilometres. How far is Holly's apartment from work? PS R

## Technology

### Fruit and vegetables

The spreadsheet below shows the items on Zoe's shopping list. Follow the steps below to complete the 'Cost' column and answer the questions.

**1** Zoe has done her fruit and vegetables shopping for the week. Enter the items into a spreadsheet, as shown below. (Centre values, bold headings, include cell borders and $ signs for column B values.)

| | A | B | C | D |
|---|---|---|---|---|
| 1 | **Item** | **Cost per kg** | **kg bought** | **Cost** |
| 2 | Oranges | $3.99 | 1 | |
| 3 | Apples | $4.99 | 0.5 | |
| 4 | Kiwi fruit | $2.99 | 0.4 | |
| 5 | Bananas | $5.60 | 1.2 | |
| 6 | Tomatoes | $7.49 | 1.5 | |
| 7 | Grapes | $3.50 | 1 | |
| 8 | Carrots | $1.55 | 2 | |
| 9 | Potatoes | $2.45 | 5 | |
| 10 | Beans | $6.95 | 0.355 | |
| 11 | Pumpkins | $3.20 | 0.562 | |
| 12 | Zucchinis | $2.85 | 0.38 | |

**2** Write a formula in cell D2 to calculate the cost of the oranges.

**3** Use **Fill Down** to calculate the cost of each item purchased.

**4** In cell C13, enter the label 'Total cost'.

**5** In cell D13, write a **sum** formula to find the cost of Zoe's shopping.

**6** In cell C14, enter the label 'Cash'. In cell C15, enter the label 'Change'.

**7** If Zoe paid $50 for her shopping, enter this value into cell D14 and in cell D15 write a formula to calculate the amount of change Zoe would have received.

**8** Zoe paid cash, but because the change was an irregular amount, she could not be given the amount in cell D15 in coins. In cell C16, enter the label 'Rounded change'. In cell D16, enter the amount of change Zoe actually received.

**9** Create a spreadsheet with similar formatting to calculate a different shopping list.

## Power plus ANSWERS ON P. 587

**1** Copy and complete this table and note the pattern.

| Fraction | $\frac{1}{7}$ | $\frac{2}{7}$ | $\frac{3}{7}$ | $\frac{4}{7}$ | $\frac{5}{7}$ | $\frac{6}{7}$ |
|---|---|---|---|---|---|---|
| Decimal | | | | | | |

**2** Investigate the value of $0.\dot{9}$.

**3** Evaluate each expression.

**a** $(0.02)^2$ **b** $(0.02)^3$ **c** $(1.1)^3$

**d** $\sqrt{0.04}$ **e** $\sqrt{0.36}$ **f** $\sqrt[3]{0.027}$

**4** Wolf, the warrior, is trying to break a decimal code that will open the dungeon doors. He has to free the prisoners before midnight so that they will not be turned into frogs by an evil spell. Each castle door is operated by a combination lock. Use the following clues to match the various combinations with the doors of the different rooms in the castle.

| Combinations | Rooms |
|---|---|
| 8.262 | Queen's chamber |
| 9.24 | armoury |
| 9.96 | throne room |
| 8.07 | banquet hall |
| 8.16 | kitchen |
| 8.79 | dungeon |

**Clues**

- Combination 9.24 opens a door to a room that deals with food.
- The combination to the armoury has a 6 in the hundredths place.
- The combinations of the throne room and the banquet hall add to 18.03.
- The Queen's chamber has a combination that is smaller than $25.11 \div 3.1$.
- The kitchen combination is one of the 3 largest combination numbers.
- The doors in the dungeon and the throne room remained locked when Wolf tried 9.96 and 8.262.

**5** Evaluate $28.98 \div 1.2$ without using a calculator.

**6** Decide where the decimal points should be so that the numbers in the ovals fit the clues.

**a** The product of the 2 numbers is 91.02.
The sum of the numbers is 28.3.

37 246

**b** The sum of the 3 numbers is 14.
The product of the 3 numbers is 78.66.

23 57 6

**c** The sum of the 4 numbers is 28.32.
The product of 2 of the numbers is 3.672.
The product of the other 2 is 81.

51 72 45 18

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# CHAPTER 7 REVIEW



## Language of maths

| | | | |
|---|---|---|---|
| ascending | decimal | decimal place | decimal point |
| descending | estimate | fraction | hundredth |
| power of 10 | recurring decimal | round down | round up |
| tenth | terminating decimal | thousandth | |

**1** What is the collective name for numbers such as 10, 100 and 1000?

**2** 'The bushfire decimated the possum population of the forest.' Look up the meaning of the word '**decimate**'.

**3** What is a **recurring** dream or a **recurring** back pain? What does 'recurring' mean?

**4** What happens when a train **terminates** at a station?

**5** What is a 'decimetre'?

**6** How many decimal places do you round to if you are rounding to the nearest hundredth?

**Quiz**
Language of maths 7

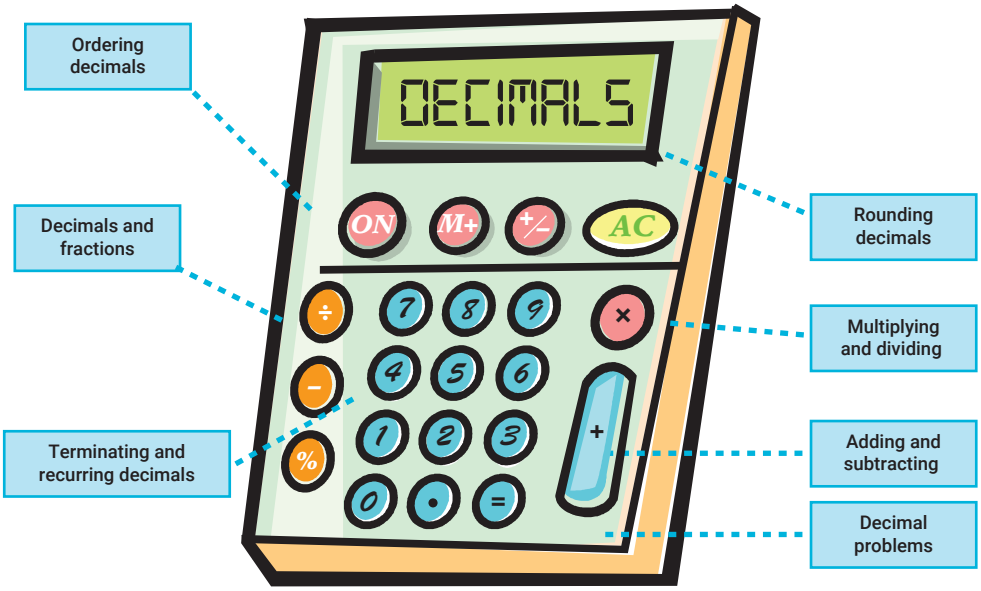
**Puzzle**
Decimals crossword

## Topic summary

- What parts of this topic were new to you? What parts did you already know?
- Write any rules you have learnt about working with decimals.
- What parts of this topic did you not understand? Be specific. Talk to a friend or your teacher about them.
- Give 3 examples of where decimals are used.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

**Worksheet**
Mind map: Decimals



Quiz
Test yourself 7

# TEST YOURSELF 7

ANSWERS ON P. 587

7.01 **1** **a** Arrange 34.98, 36.86, 3.998, 38.141, 340, 34.89 in ascending order.

**b** Arrange 1.015, 1.293, 1.1015, 1.239, 1.006 in descending order.

7.02 **2** Convert each decimal to a simplified fraction.

**a** 0.5 **b** 0.89 **c** 0.09 **d** 0.444

7.02 **3** Convert each fraction to a decimal.

**a** $\frac{4}{10}$ **b** $\frac{13}{100}$ **c** $\frac{7}{100}$ **d** $\frac{11}{1000}$

7.03 **4** Evaluate each expression.

**a** $12.35 + 4.53 + 0.56 + 3.125 + 24.7 + 20.09$

**b** $214.33 - 109.84$

**c** $0.568 + 23 + 4.027 - 16.28$

**d** $1600.8 - 562.9$

**e** $1453.6 + 1287.31 - 2344.4$

**f** $9.23 - 6.851$

7.04 **5** Evaluate each expression.

**a** $7.54 \times 10$ **b** $7.54 \times 100$ **c** $7.54 \times 1000$

**d** $13.9 \div 10$ **e** $13.9 \div 100$ **f** $13.9 \div 1000$

7.05 **6** Given that $42 \div 76 = 3192$, find:

**a** $4.2 \times 76$ **b** $4.2 \times 7.6$ **c** $0.42 \times 760$

**d** $4.2 \times 0.76$ **e** $42 \times 7.6$ **f** $0.42 \times 0.76$

7.06 **7** Evaluate each product.

**a** $2.75 \times 6$ **b** $0.5 \times 1.2$ **c** $72.23 \times 4$

**d** $6.1 \times 1.2$ **e** $0.92 \times 5$ **f** $3 \times 9.7$

**g** $3.25 \times 0.41$ **h** $0.05 \times 0.02$ **i** $4.67 \times 1.1$

7.07 **8** Evaluate each quotient.

**a** $762.4 \div 2$ **b** $97.6 \div 8$ **c** $2.75 \div 4$ **d** $195.6 \div 3$

7.08 **9** Evaluate each quotient.

**a** $12.5 \div 0.5$ **b** $12.72 \div 0.4$ **c** $6.9 \div 0.03$ **d** $0.508 \div 0.02$

7.09 **10** Convert each fraction to a decimal.

**a** $\frac{4}{5}$ **b** $\frac{3}{8}$ **c** $\frac{5}{9}$ **d** $\frac{2}{3}$

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**11** Round: 7.10

- **a** 406.28 to the nearest tenth
- **b** 125.724 to 2 decimal places
- **c** 2345.876 to 1 decimal place
- **d** 3.8967 to the nearest hundredth
- **e** 78 654.056 to 2 decimal places
- **f** 678.4309 to the nearest thousandth

**12** The Liverpool Women's Cricket Club is having a pizza night. They order 16 Super Supreme pizzas at $19.70 each and 10 Hawaiian pizzas at $15.50 each. How much will the club spend on pizzas? 7.11



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**13** Riley bought 825 bricks for $1105.50. How much did one brick cost? 7.11

**14** Mia saved $220 to go to a rock concert. Her return fare cost $18.80, her concert ticket cost $116.75, the program cost $25 and food cost $37.70. She did not have enough to buy the band's backpack (priced $59.95) after the concert. How much did she need to borrow from her friend Sam to buy the backpack?